

Not All Cops Are Bayesian: Selection Neglect in Policing Decisions

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Motivation

- Significant **racial and class policing disparities** —stops, arrests, use of force, etc.
 - Recently shown by Pierson et al. (2020), Chen et al. (2023), Aggarwal et al. (2025), Feigenberg and Miller (2025), Hoekstra and Sloan (2022), Goncalves and Mello (2021), Vomfell and Stewart (2021), Ba et al. (2021), Rizzotti (2024), Bor et al. (2018), and Gonçalves, Mello, and Weisburst (2025) and others

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- Are disparities driven by **taste** or by **beliefs**?
 - Literature tests whether disparities can be explained by *accurate statistical discrimination*
 - When observed disparities deviate from Bayesian inference, behavior is attributed to taste

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- Are disparities driven by **taste** or by **beliefs**?
 - Literature tests whether disparities can be explained by *accurate statistical discrimination*
 - When observed disparities deviate from Bayesian inference, behavior is attributed to taste
 - But if most people are not Bayesian, **why assuming police officers are?**

Selection Bias in Data-Driven Policing

Police officers make **crime predictions** based on **crime data** (crime reports, arrests) (Brayne 2020)

- Crime Data suffers from **selection bias**:
 - More stops of Black people → More common in crime data (Gonçalves, Mello, and Weisburst 2025)

If officers **neglect selection** bias in crime data → **biased data-driven policing**

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Does selection neglect generate policing disparities even in the absence of taste bias?

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Why an Experiment?

- Taste-free mirror of policing decisions + control available data
- Clear objective of patrolling decisions (Stashko 2023)

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- Data used for crime predictions is **reported crime**
- Every day, station commanders receive geographic distribution of reports and **decide where to send patrols**
- Report data has **selection bias** → **29% of crimes in Colombia are reported** (ECSC 2023)
 - Varies by crime: 7% of extortions vs. ~ 100% homicides are reported
 - Varies by area: 45% reported in Bogotá, 16% in rural areas

Police in Colombia uses reported crime data to make predictions, selected through reporting

Experimental Design

Patrolling Task

Taste-Free Mirror of Policing Decisions

Generate fictitious crime data about 2 cities $g \in \{A, B\}$ — placeholder for race, class...

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- Show demographics and reported crime data to make prediction.
- Reported crime data is **selected through reporting rate**

Patrolling Task

Decision Screen

City A		City B	
Zone 2 out of 10		Zone 2 out of 10	
Reports	60	Reports	56
Reporting Rate	35%	Reporting Rate	30%
Average Age	32	Average Age	26
Unemployment	18%	Unemployment	13%
Socioeconomic Status	4	Socioeconomic Status	4

Click on a zone to patrol it

Calculator

Patrolling Task

Data Generating Process

To infer $\hat{c}_{gn}$, participants can use:

- **Reporting Rate** (α_{gn}) $\rightarrow$ probability of a crime being reported, $\alpha_{gn} \sim U[0.1, 0.9]$
 - Determine data selection: $\alpha_A \neq \alpha_B \implies$ **selection bias**

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Officers know that **sufficient statistic is adjusted reports** $\hat{c}_{gn} = r_{gn}/\alpha_{gn}$

If 40 reports from an area with 50% reporting rate, best estimate is 80 crimes

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- **Demographic Data** $\rightarrow$ average age, unemployment rate, and SES (1-6)
 - All independent and uniformly distributed, uninformative of crime

Experiment Structure

Block 1

- Participants make 9 **patrolling decisions**
 - See reports, reporting rate, age, unemployment and SES
- For 10th decision, they see reports + choose 1 other data → measure of **information choice**
- Record patrolling choices and response times, max 1 minute

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Block 2

- Participants make 12 patrolling decisions
 - Only see reports and reporting rate → selection is made more salient
- Record patrolling choices and response times, max 1 minute

Participants and Procedures

- Recruit 182 **Captains from the National Police of Colombia**
 - Training in National Academy to promote to Majors → **entire cohort participates**
 - 15+ years of experience + at least college education
 - Many acting as station commanders → **experience making patrol decisions**
 - “We want to learn from experts how patrolling decisions are made”
- Multiple questions testing **understanding of the instructions** and **math skills**
 - We take mistakes as a measure of confusion → all results are robust
- On average, paid US \$10.5 for 40 minutes of experiment

Do Police Officers **Adjust for Data Selection?**

Selection Bias in Crime Data

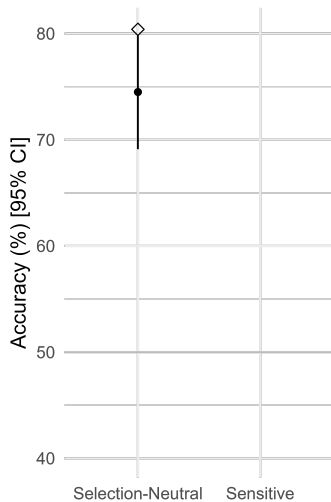
We classify decisions according to whether optimal decision is sensitive to selection bias:

- **Selection-Neutral Decisions** → adjusting for selection **doesn't change** optimal decision
 - $\text{sign}(\Delta \text{Reports}) = \text{sign}(\Delta \text{Crimes})$
- **Selection-Sensitive Decisions** → adjusting for selection **changes** optimal decision
 - $\text{sign}(\Delta \text{Reports}) \neq \text{sign}(\Delta \text{Crimes})$

Accurate Decision → Patrol neighborhood with most crimes

More errors in selection-sensitive decisions $\implies$ **Selection Neglect**

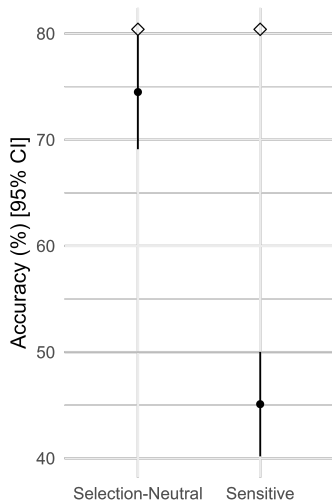
Selection Neglect



Diamonds show Bayesian Benchmark

Selection Neglect

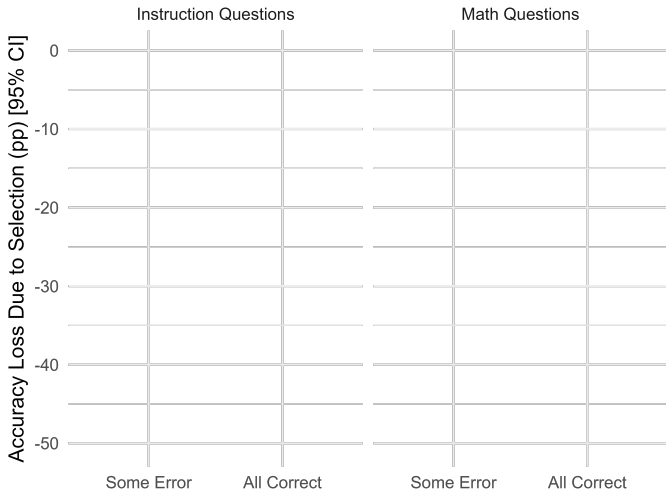
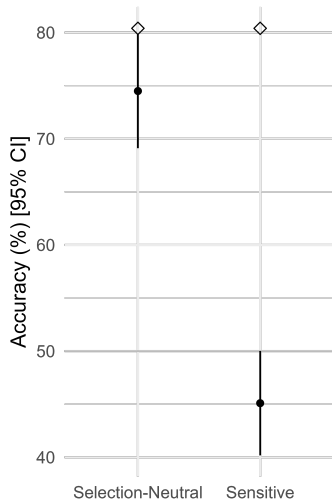
~ 30pp Accuracy Loss Due to Selection



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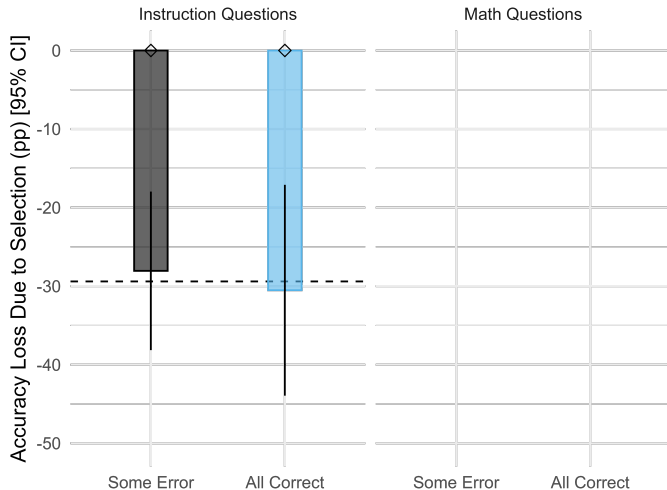
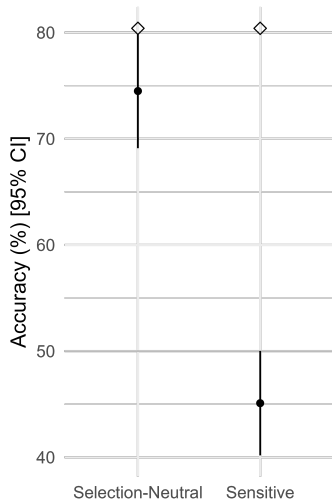
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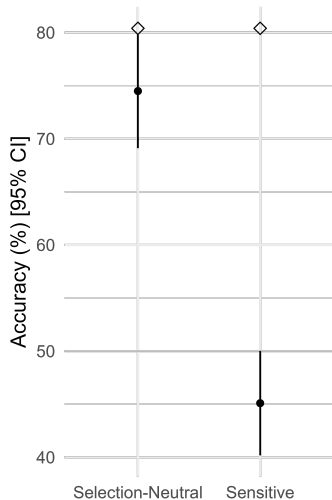
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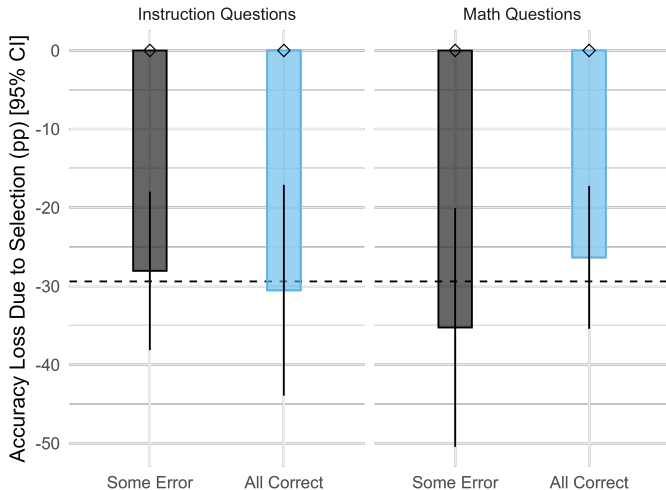
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Robust to instructions understanding and math ability



Diamonds show Bayesian Benchmark

What Drives **Selection Neglect**?

What would Bayes Do?

(1) Specifying the *right* mental model

- Officers might not be looking at data selection at all → **extensive margin**
- Data selection just doesn't come to mind → depends on salience (Enke 2020; Bordalo et al. 2023)

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(2) Solving the problem *correctly*

- Even when looking at data selection, officers might not adjust enough → **intensive margin**
- More of a computational constraint story (Enke et al. 2024; Arrieta and Nielsen 2024)

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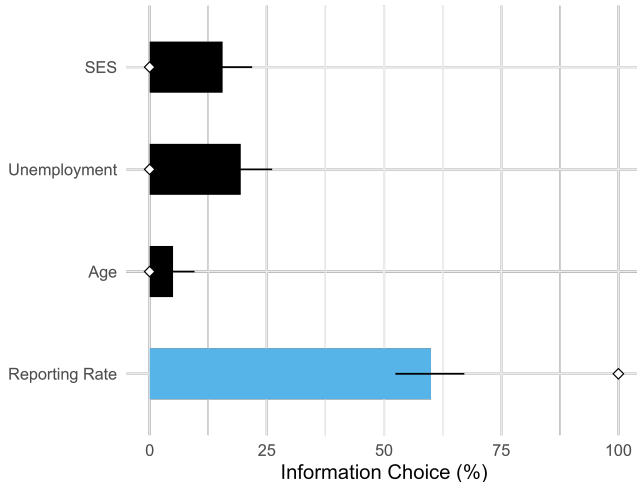
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Before last decision, we ask officers to **choose 1 data to see beyond reports**

- See Reports + (Reporting Rate/Age/Unemployment/SES)
- Informative about their mental model

Information Choice

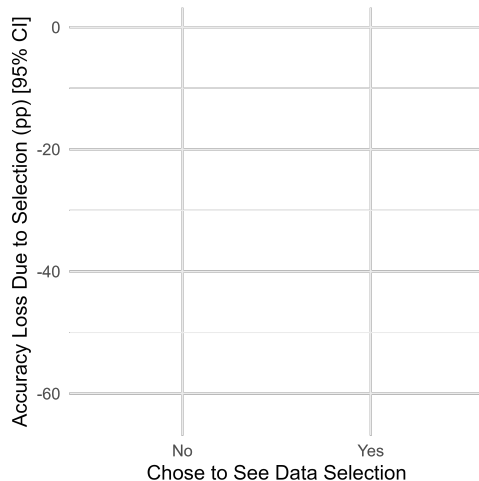
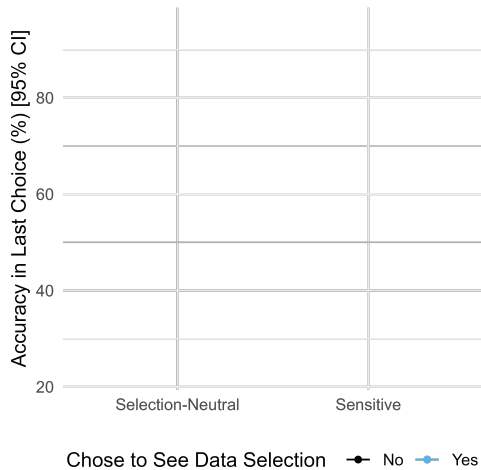
What Data do Police Officers Choose to See for Last Decision?



40% of officers prefer **not to**
see data selection →

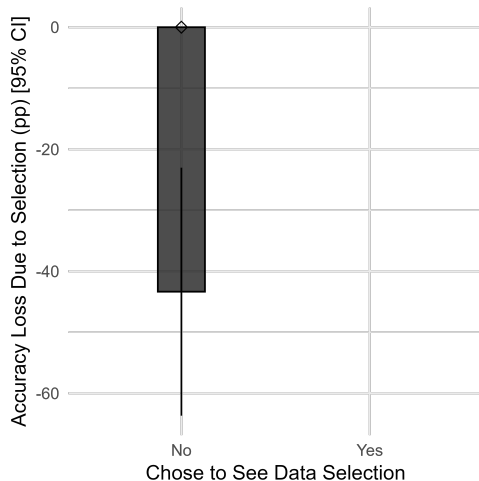
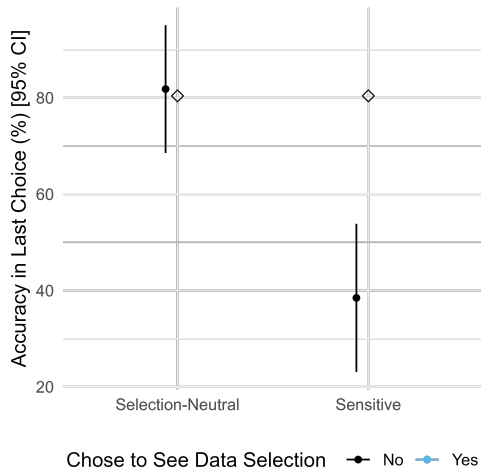
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Selection Neglect at Intensive Margin



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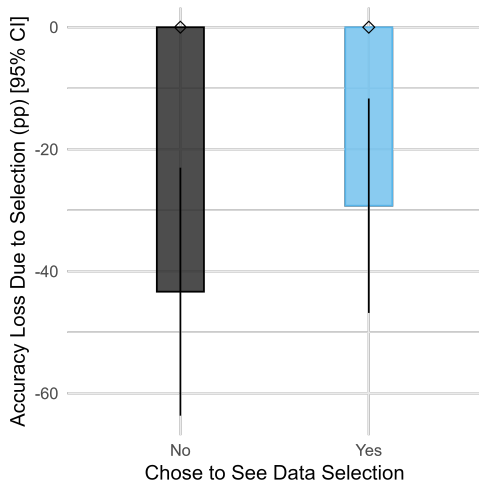
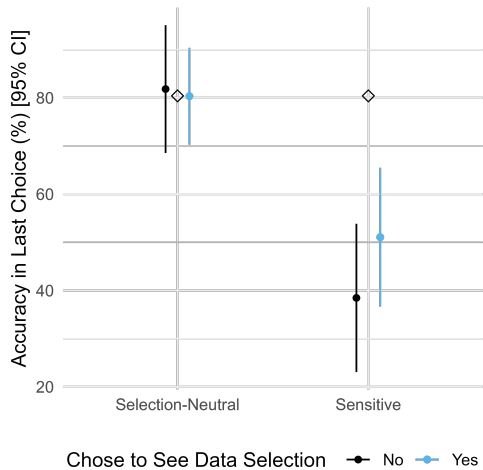
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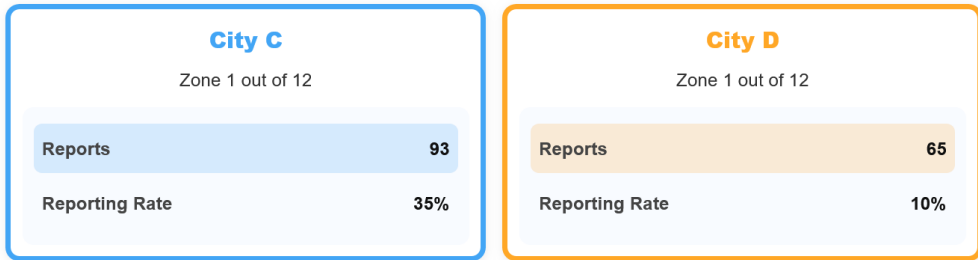
Conditional on looking at data selection, officers still **don't adjust** by it



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Block 2

Making Crime Data Selection More Salient

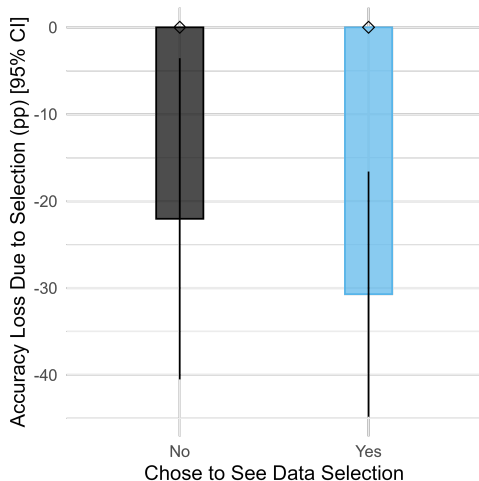
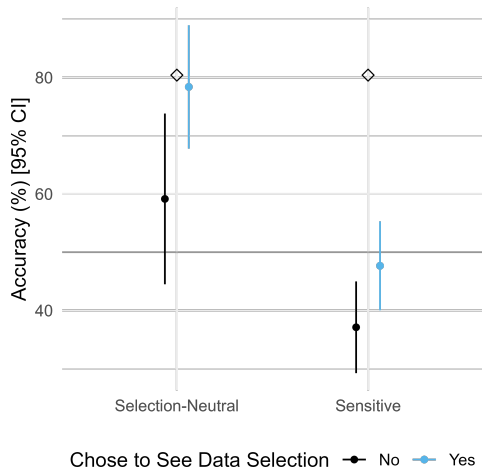


Click on a zone to patrol it

Calculator

Salience doesn't reduce Neglect

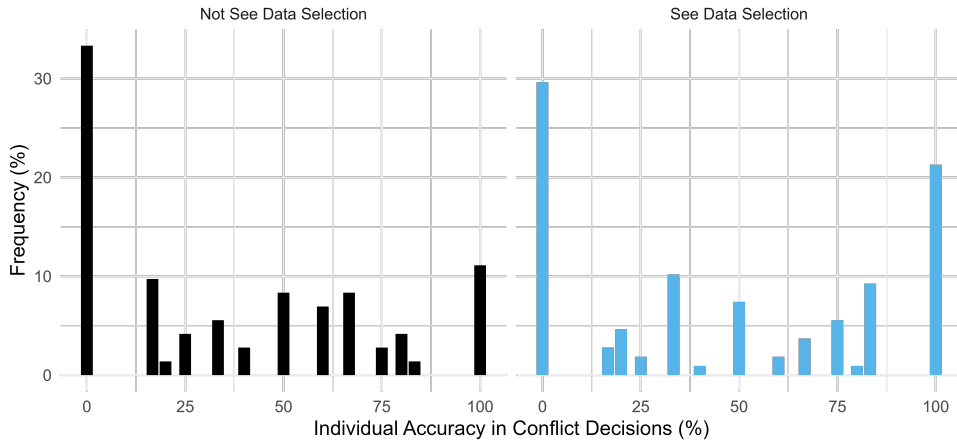
Officers still neglect selection even when made more salient



Diamonds show Bayesian Benchmark. Data only includes 6 first decisions of Block 2.

Bimodality in Neglect

Only 17% of officers fully adjust for selection



Data only includes 6 first decisions of Block 2

Heterogeneity

We elicit officers' trust in reported crime data:

- (1) Crime reports capture the geographical distribution of actual crime
- (2) Crime reports capture the time trends in actual crime
- (3) Citizens know well how and where to report a crime
- (4) Crime reports are the best data available to track crime
- (5) Crime reports provide a biased picture of actual crime

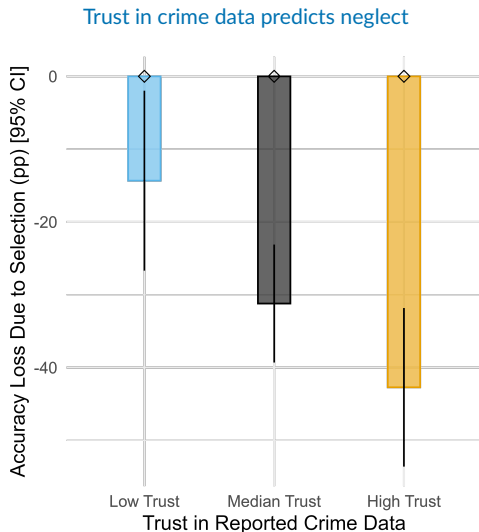
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Selection Neglect in Policing Decisions

We find **robust evidence of selection neglect in policing decisions**:

- Extensive Margin → 40% of officers choose not to see data selection
 - Trust in reported crime data predicts neglect
- Intensive Margin → ~80% who choose to see data selection don't adjust enough by it
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Not All Cops Are Bayesian

Empirical tests of taste-based discrimination need to consider *inaccurate* beliefs as H_0

Consequences of Selection Neglect

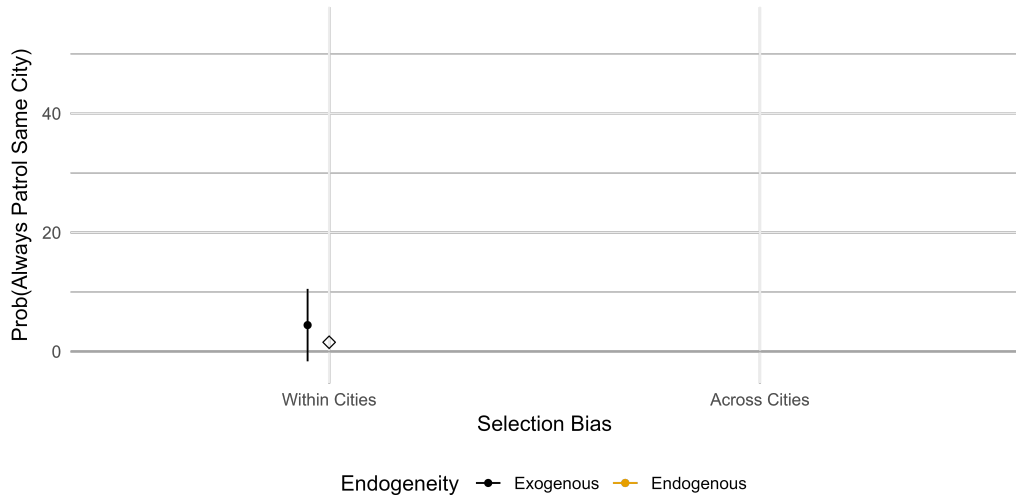
Treatments in Block 2

So far, we have studied selection bias within cities (trial-by-trial). What happens when there is systematic selection bias across groups? Last 6 decisions of Block 2, randomize selection bias:

- **Within Cities** → reporting rates for both cities drawn from same distribution
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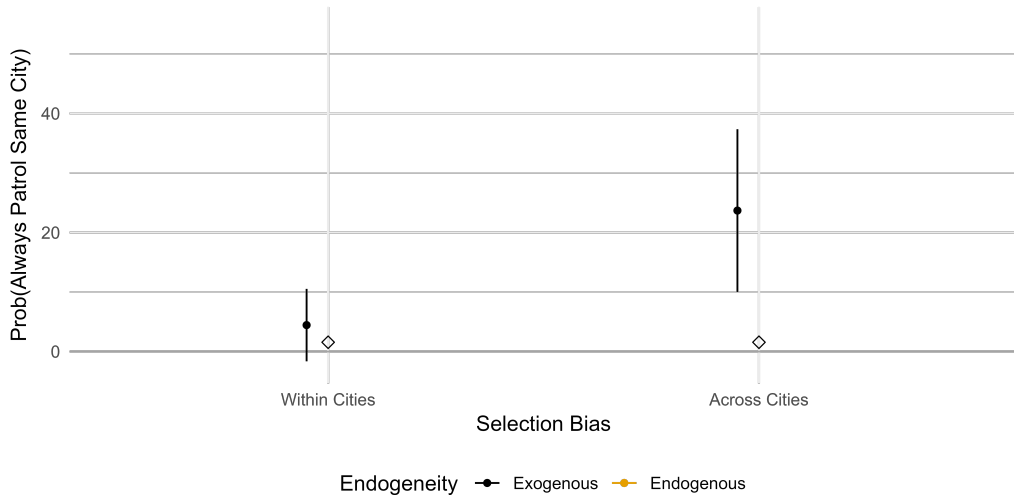
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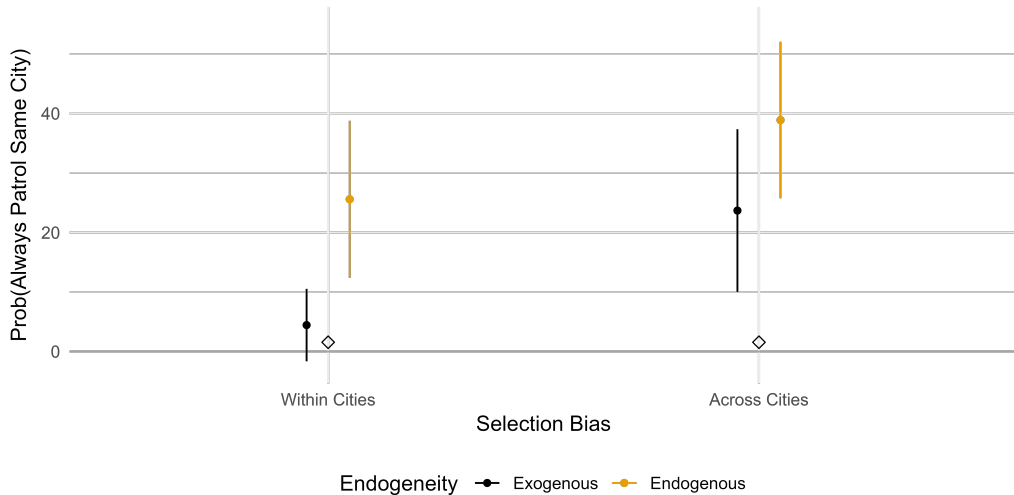
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What happens when patrolling decisions affect selection? x2 treatments

- **Exogenous Selection** → patrols don't change reporting rates
- **Endogenous Selection** → patrolling one city increases reporting rates there by +15% in the next decision

Overpolicing

Systematic Selection Bias and Endogenous Selection → Overpolicing



Selection Neglect and Policing Disparities

Selection neglect causes policing disparities: biased data → biased data-driven policing

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Not All Cops Are Bayesian

We mirror observed policing disparities, in the absence of taste-based discrimination

How to **reduce selection neglect** in policing?

Predictive Policing Algorithms as Debiasing Tools?

Predictive policing algorithms “do the math” for officers, but:

- Evidence of selection neglect in predictive algorithms (Lum and Isaac 2016)
 - Can amplify cycles of bias *by design*

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We test the effect of predictive policing algorithms on policing decisions

Experimental Design

We use an algorithm that predicts crime from reported data, using past data with decaying weights (hot-spot), and makes a **patrol recommendation**.

Experimental Design

City C

Zone 8 out of 12

Reports	29
---------	----

Reporting Rate	10%
----------------	-----

Recommended by algorithm

City D

Zone 8 out of 12

Reports	232
---------	-----

Reporting Rate	90%
----------------	-----

Click on a zone to patrol it

Experimental Design

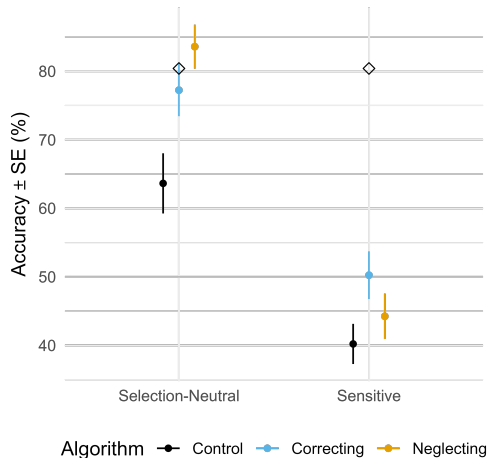
We use an algorithm that predicts crime from reported data, using past data with decaying weights (hot-spot), and makes a **patrol recommendation**.

For last 6 patrolling decisions of Block 2, we randomize officers to 3 groups:

- **Correcting Algorithm** → Adjusts for data selection
 - Improve all decisions
- **Neglecting Algorithm** → Doesn't adjust for data selection
 - Improve selection-neutral decisions and worsen sensitive decisions
- **Control** → No recommendation

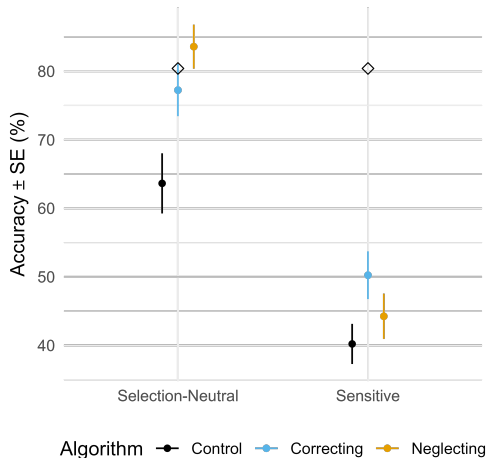
Do Algorithms Improve Decisions?

Only when selection doesn't bias decision

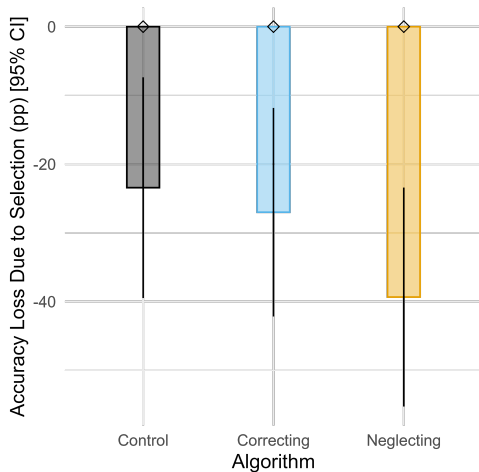


Do Algorithms Improve Decisions?

Only when selection doesn't bias decision



No reduction of selection neglect



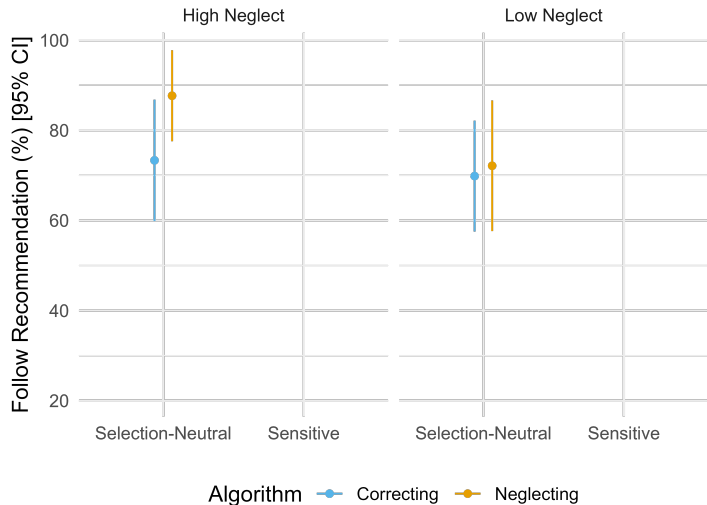
Who Follows the Algorithm?

Predictive policing algorithms can be **ineffective** if their recommendations are only followed when they confirm priors.

Divide the sample in 2: above and below the median accuracy in selection-sensitive decisions (33%). If there is confirmation bias, we expect:

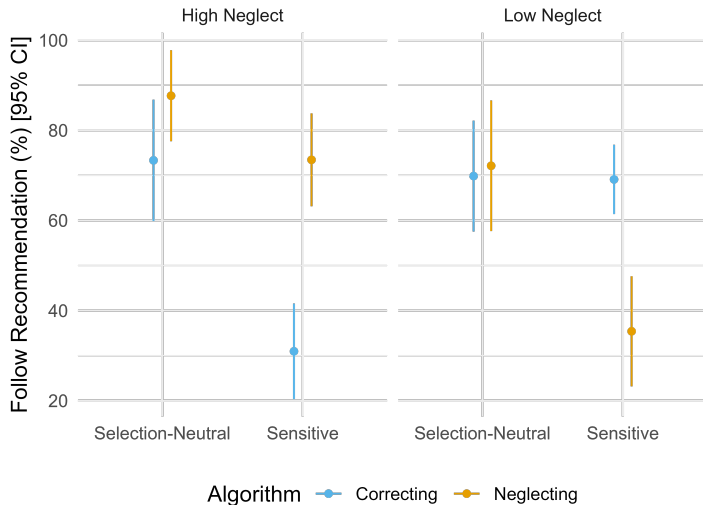
- **Low Neglect** → follow correcting > neglecting algorithm in selection-sensitive decisions
- **High Neglect** → follow neglecting > correcting algorithm in selection-sensitive decisions

Who Follows the Algorithm?



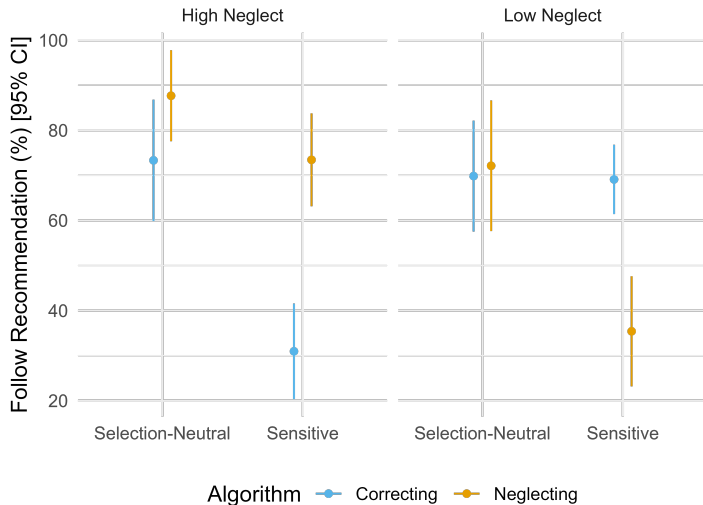
Who Follows the Algorithm?

Officers follow when confirms priors

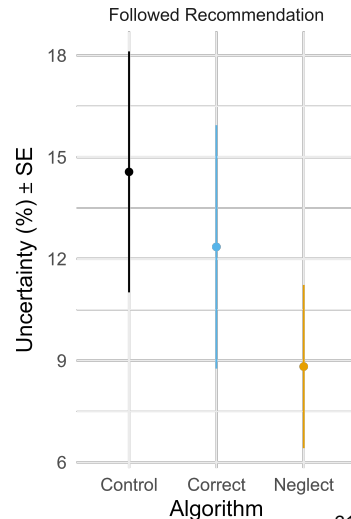


Who Follows the Algorithm?

Officers follow when confirms priors



Reducing uncertainty (underpowered)



Conclusions

Not All Cops Are Bayesian

Police officers **neglect the selection bias** prevalent in crime data. As a consequence:

- (1) Systematic data selection leads to **policing disparities in the absence of taste**
- (2) Tests of taste-based discrimination should consider **inaccurate beliefs** as H_0
- (3) Predictive policing algorithms **don't correct bias**, but might **amplify and justify** it

Thank you so much!

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Discrete Choice

Using decisions *before* information choice, estimate discrete choice model:

$$P(patrol_A) = \beta_1 \Delta(Reports) + \beta_2 \Delta(Adj.Reports) + \beta_3 \Delta(Age) + \beta_4 \Delta(Unemployment) + \beta_5 \Delta(SES) + \epsilon \quad (1)$$

where we define $\Delta(Adj.Reports) = \frac{Reports_A}{ReportRate_A} - \frac{Reports_B}{ReportRate_B}$

Discrete Choice

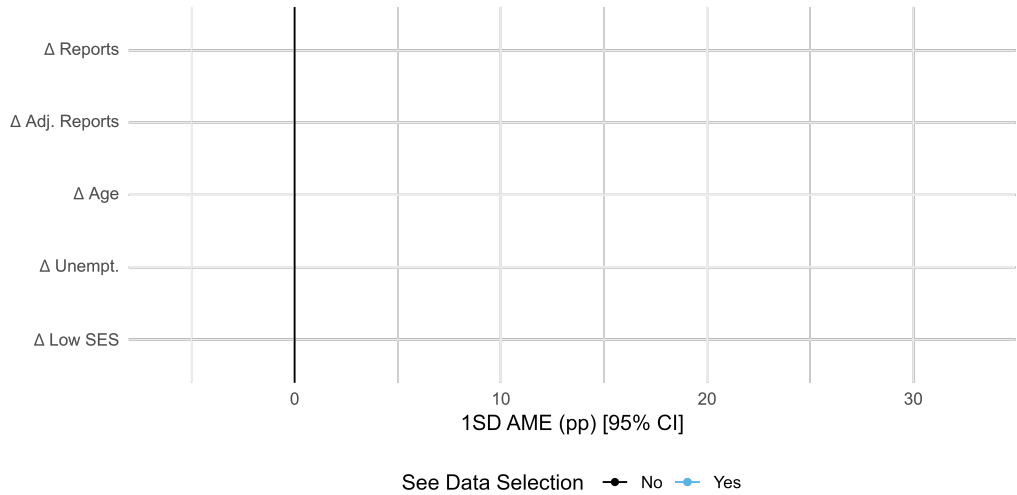
Using decisions *before* information choice, estimate discrete choice model:

$$P(\text{patrol}_A) = \beta_1 \Delta(\text{Reports}) + \beta_2 \Delta(\text{Adj. Reports}) + \beta_3 \Delta(\text{Age}) + \beta_4 \Delta(\text{Unemployment}) + \beta_5 \Delta(\text{SES}) + \epsilon \quad (1)$$

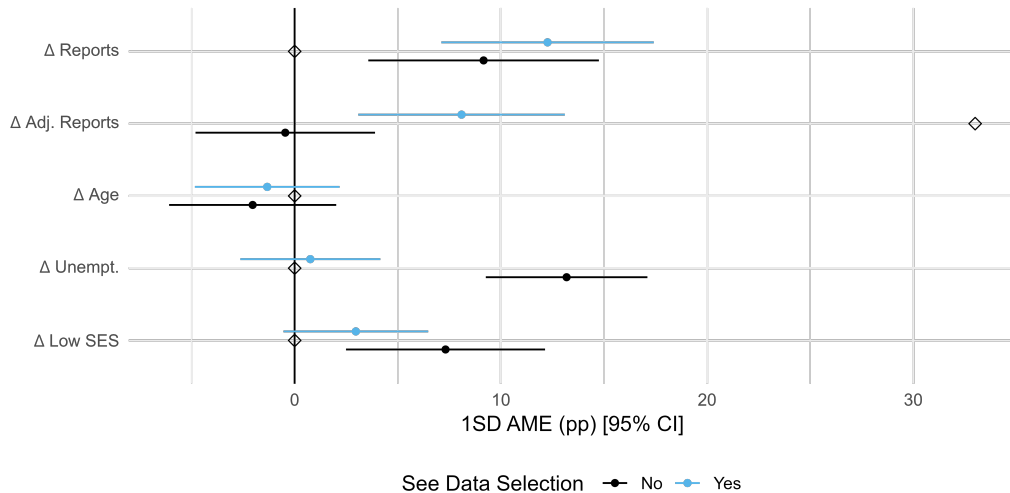
where we define $\Delta(\text{Adj. Reports}) = \frac{\text{Reports}_A}{\text{ReportRate}_A} - \frac{\text{Reports}_B}{\text{ReportRate}_B}$

- Bayesian Benchmark → **Only use Adjusted Reports**
 - Conditional on Adj. Reports, variation in Reports only comes from selection.
- Plot the **1SD Average Marginal Effect** on $P(\text{patrol})$
 - How much does the probability of patrol change when increasing a predictor by 1SD?

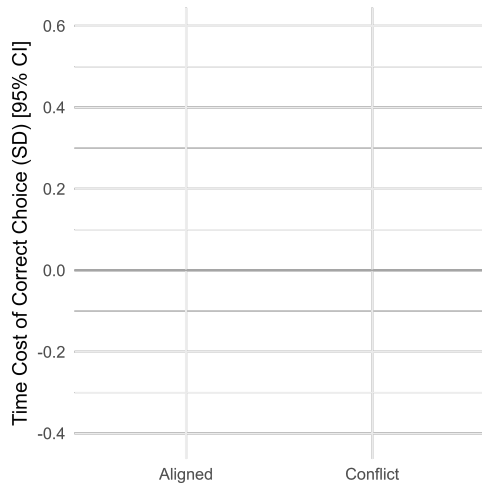
Discrete Choice



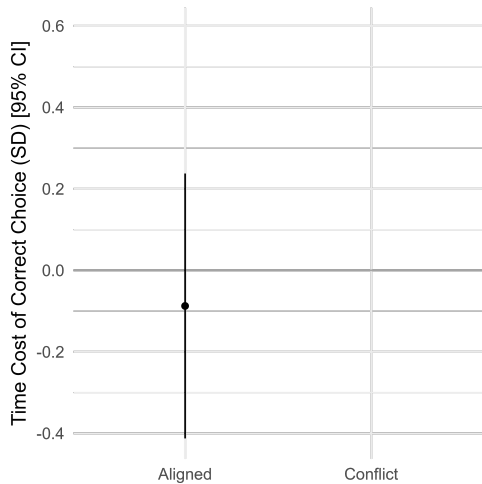
Discrete Choice



Cognitive Cost

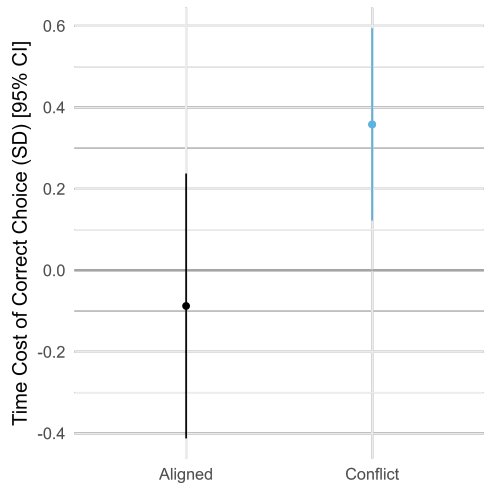


Cognitive Cost



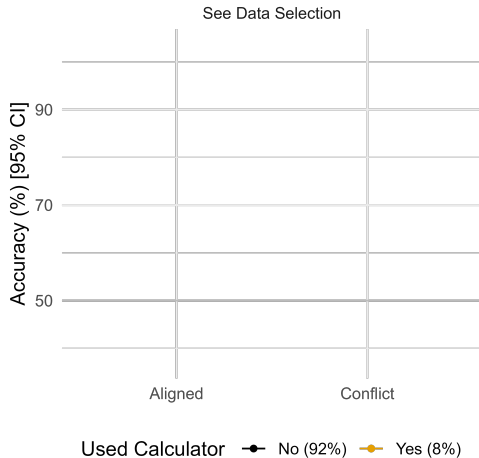
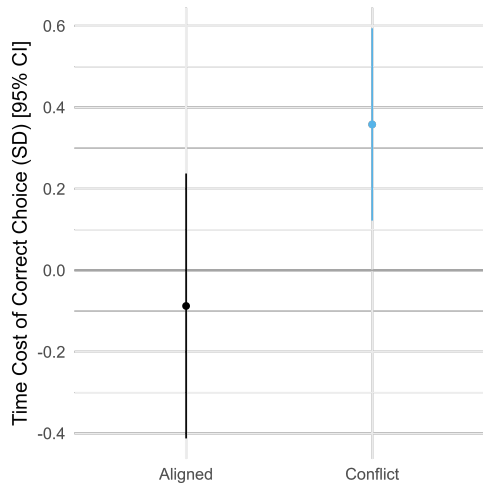
Cognitive Cost

Correcting selection has cognitive cost



Cognitive Cost

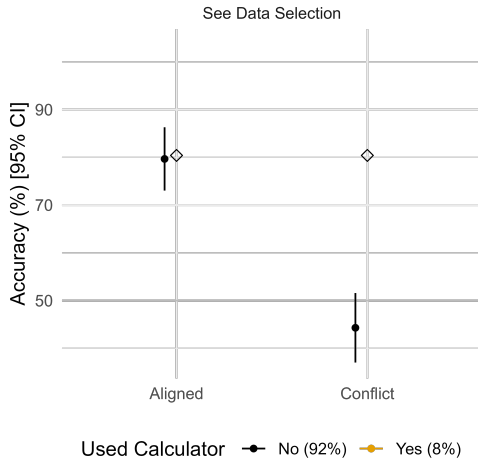
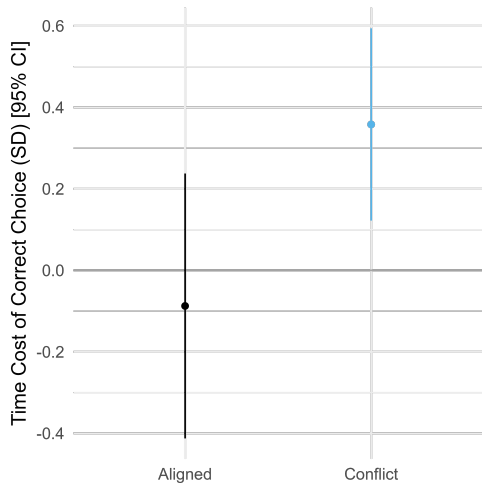
Correcting selection has cognitive cost



Diamonds show Bayesian Benchmark

Cognitive Cost

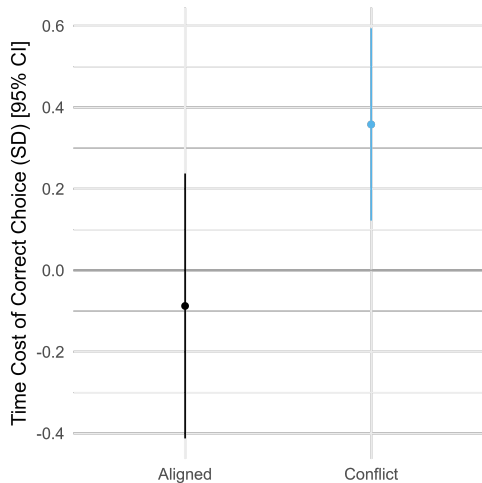
Correcting selection has cognitive cost



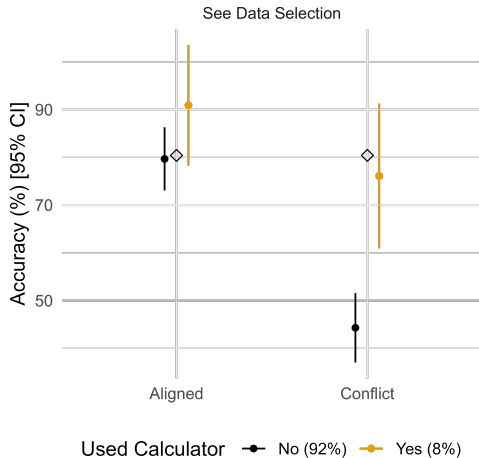
Diamonds show Bayesian Benchmark

Cognitive Cost

Correcting selection has cognitive cost



Calculator closes the gap



Diamonds show Bayesian Benchmark